CANTEEN ORDER SCHEDULING SYSTEM

DAA Mini Project • Python 3

1. Introduction

A canteen receives multiple food orders with different preparation times and priorities. The order in which jobs are prepared directly affects customer waiting time. This project models a single preparation counter and compares three scheduling strategies.

2. Problem Statement

Design a scheduling system that arranges canteen orders using different algorithmic rules and measures waiting time and turnaround time so that scheduling performance can be compared objectively.

3. Objectives

• Represent orders using ID, food item, preparation time and priority.
• Implement FCFS, Shortest Processing Time and Priority Scheduling.
• Calculate start, completion, waiting and turnaround times.
• Compare algorithms using measurable metrics.
• Demonstrate correctness and complexity.

4. Algorithm Design

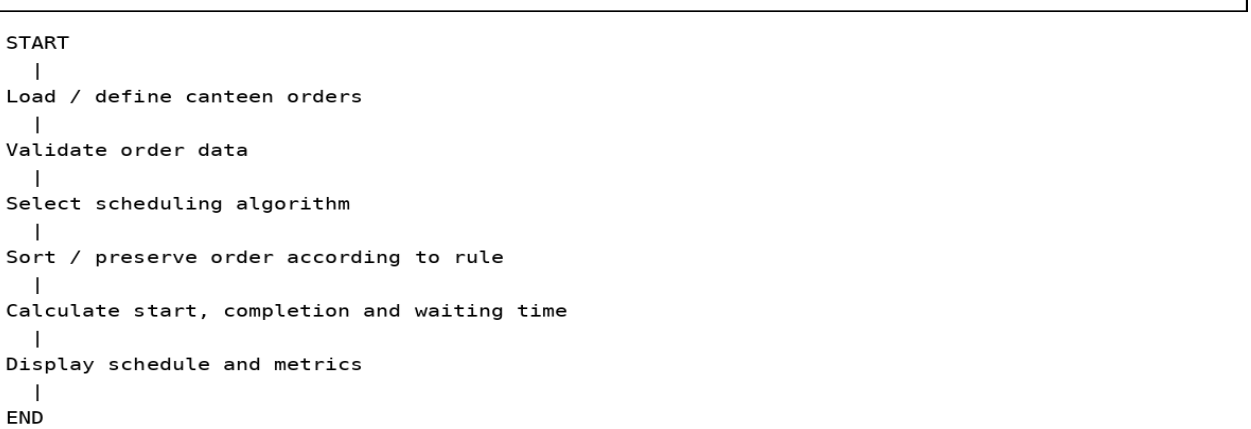
FCFS: preserves the input/arrival order.
SPT: sorts by ascending preparation time. With one counter and all jobs available initially, it minimizes average waiting time.
Priority: sorts by descending priority; ties use shorter preparation time and order ID.

5. Correctness

Each algorithm schedules every input order exactly once. FCFS preserves sequence, SPT creates nondecreasing preparation times, and Priority creates nonincreasing priorities with deterministic tie-breaking. Completion time is accumulated from the previous job, so waiting and turnaround values follow the generated schedule.

6. Flowchart

System Flowchart



7. Technologies / Programming Language Used

Python 3, Python standard library, unittest for testing, and ReportLab for project documentation.

8. Complexity Analysis

Algorithm	Time Complexity	Space Complexity	Main Property
FCFS	O(n)	O(n)	Simple arrival order
SPT	O(n log n)	O(n)	Lowest average waiting under standard assumptions
Priority	O(n log n)	O(n)	Higher priority served first

Sorting dominates SPT and Priority. The scheduling pass is linear. The returned schedule requires O(n) space.

9. Test Cases and Output Screenshots

Sample orders: O1 Thali (12 min, priority 2), O2 Sandwich (5 min, priority 1), O3 Dosa (8 min, priority 3), O4 Tea (3 min, priority 1).

```
Program Menu
```

```
CANTEEN ORDER SCHEDULING SYSTEM
1. Show Orders
2. FCFS Scheduling
3. Shortest Processing Time (SPT)
4. Priority Scheduling
5. Compare Algorithms
6. Exit
Enter choice: 5
```

```
SPT Result
```

```
SPT SCHEDULING RESULT
Order   Item              Start   Finish   Wait
O4      Tea               0       3        0
O2      Sandwich          3       8        3
O3      Dosa              8       16       8
O1      Thali             16      28       16
Average waiting time: 6.75 min
```

10. Output and Performance Comparison

```
Performance Comparison
```

```
ALGORITHM COMPARISON
Algorithm       Avg Waiting   Avg Turnaround   Total Time
FCFS              14.25          21.25            28
SPT                6.75          13.75            28
Priority           9.50          16.50            28
SPT gives the lowest average waiting time for this sample.
```

Algorithm	Average Waiting	Average Turnaround	Total Time
FCFS	14.25 min	21.25 min	28 min
SPT	6.75 min	13.75 min	28 min
Priority	9.50 min	16.50 min	28 min

For this sample, SPT produces the lowest average waiting time because short preparation jobs are completed before longer jobs. Priority scheduling is useful when business importance must override pure waiting-time minimization.

11. Project Structure

```
Canteen-Order-Scheduling-System/
■■■ canteen_scheduler.py
■■■ README.md
■■■ requirements.txt
■■■ algorithms/ (fcfs.py, spt.py, priority.py)
■■■ utils/helpers.py
■■■ tests/test_scheduling.py
■■■ docs/sample_output.txt
■■■ screenshots/
```

12. Execution and Testing

Run `python canteen_scheduler.py`. Run tests with `python -m unittest discover -s tests -v`. The test suite checks FCFS ordering, SPT ordering and average waiting time, and Priority ordering.

Expected test result: 3 tests pass successfully.

13. Conclusion

The project demonstrates how algorithm selection affects a practical canteen scheduling problem. FCFS is straightforward, SPT is effective for reducing average waiting time under standard assumptions, and Priority Scheduling handles cases where some orders deserve earlier service.

14. Future Scope

Future versions can support multiple counters, real-time order arrivals, cancellations, kitchen capacity, customer notifications, dynamic priorities, and a web/mobile interface.

15. Submission Checklist

GitHub repository link • Source code • Project Report PDF • README with execution instructions • Test cases • Output screenshots.

Suggested GitHub repository name: `Canteen-Order-Scheduling-System`.